

# Lucas Howard

POSTDOCTORAL ASSOCIATE

Computing and Data Science, Boston University

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## Education

### University of Colorado

PHD ATMOSPHERIC AND OCEANIC SCIENCE

- Advisor: Professor Aneesh Subramanian
- Dissertation: **Advancing Earth System Prediction and Data Assimilation Using Probabilistic Machine Learning**

Boulder, CO

2021-2026

### University of Vermont

MS CIVIL AND ENVIRONMENTAL ENGINEERING

- Advisor: Professor Donna Rizzo
- Thesis: "Leveraging the Information Content of Process Based Models Using Differential Evolution and the Extended Kalman Filter"

Burlington, VT

2014-2016

### Reed College

BA PHYSICS

- Thesis Advisor: Professor Joel Franklin
- Thesis: "A Numerical Investigation of Water Waves"
- Extracurricular: Senior Reactor Operator at the on-campus nuclear research reactor

Portland, OR

2009-2013

## Appointments

### Boston University

POSTDOCTORAL ASSOCIATE

- PI: Professor Elizabeth Barnes

Boston, MA

2026-present

## Honors and Awards

2025 **Weather Innovation for Next Generation Scientists (WINGS) Fellowship**

NOAA/UCAR CPAESS

2025 **Department Service Award** For mentoring an REU student

ATOC/CU Boulder

2025 **ATOC Fellowship** Travel award (\$2,300)

ATOC/CU Boulder

2016 **Hydrogeologic Research and Development Fund** (\$13,000)

University of Vermont

## Refereed Publications and Proceedings

### JOURNAL ARTICLES

**Howard, L. J.**, Subramanian, A. C., Nadimpali, J. R., Giglio, D., & Hoteit, I. (2026). Skillful Subseasonal Marine Heat Wave Forecasts using a Neural Network *Environmental Data Science*. 5, e6. <https://doi.org/10.1017/eds.2026.10033>

**Howard, L. J.**, Subramanian, A. C., Thompson, G., Johnson, B., & Auligne, T. (2026). Probabilistic Emulation of the Community Radiative Transfer Model Using Machine Learning. *Artificial Intelligence for the Earth Systems*. 5, 240109. <https://doi.org/10.1175/AIES-D-24-0109.1>

**Howard, L. J.**, Subramanian, A., & Hoteit, I. (2024). A Machine Learning Augmented Data Assimilation Method for High-Resolution Observations. *Journal of Advances in Modeling Earth Systems*, 16(1), e2023MS003774. <https://doi.org/10.1029/2023MS003774>

Ozbek, M., Voorhies, N., **Howard, L.**, Swanson, R., & Fox, T. (2024). Delineation of a PFOA Plume and Assessment of Data Gaps in its Conceptual Model Using PlumeSeeker™. *Groundwater*, 62(1), 44–59. <https://doi.org/10.1111/gwat.13373>

**Howard, L. J.**, Anderson, I. A., Underwood, K. L., Dewoolkar, M. M., Deschaine, L. M., & Rizzo, D. M. (2016). Heuristic assessment of bridge scour sensitivity using differential evolution: case study for linking floodplain encroachment and bridge scour. *Environmental Systems Research*, 5(1), 1–12.

Akimana, R. M., Bista, H., Seo, Y., Li, L., **Howard, L. J.**, Dewoolkar, M. M., & Hu, L.-B. (2016). Multi-scale experimental and numerical study of microbially-induced calcite precipitation in sandy soils: preliminary evidence and observations. In *Geo-China 2016* (pp. 77–84).

Akimana, R. M., Seo, Y., Li, L., **Howard, L. J.**, Dewoolkar, M. M., & Hu, L. B. (2016). Exploring X-ray computed tomography characterization and reactive transport modelling of microbially-induced calcite precipitation in sandy soils. In *Geo-Chicago 2016* (pp. 62–71).

## CONFERENCE PAPERS

**Howard, L.**, Ross, J., Jarrett, M., and Amos, M. (2020). *Uncertainty quantification and calibration of a large subsurface flow and transport model using the ensemble Kalman filter*. Waste Management Symposium, Phoenix, AZ. March 10-14.

## Presentations and Posters

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\* Invited talk

March 12, 2026. *Improved Utilization of Dense Observations in Data Assimilation Systems Using Machine Learning*. Presentation. NOAA Library Webinar.

\*September 15, 2025. *Augmenting Ensemble Data Assimilation with Machine Learning for High Resolution Observations*. Presentation. Australian Bureau of Meteorology Annual R&D Workshop. Melbourne, Australia.

April 18, 2025. *Probabilistic Emulation of the Community Radiative Transfer Model Using Machine Learning*. Presentation. ATOC Colloquium. Boulder, CO.

\*April 15, 2025. *Probabilistic Emulation of the Community Radiative Transfer Model Using Machine Learning*. Presentation. Taiwan-US Workshop on AI/ML for Satellite Data, Severe Weather, and Tropical Cyclones. Boulder, CO.

\*April 1, 2025. *Machine Learning Approaches for Assimilation of High-Resolution Observations*. Presentation. Applied Mathematics Geosciences Seminar. Boulder, CO.

January 14, 2025. *Probabilistic Emulation of the Community Radiative Transfer Model Using Machine Learning*. Presentation. Annual Meeting of the American Meteorological Society, New Orleans, LA.

February 21, 2024. *A Machine Learning Augmented Data Assimilation Method for High-Resolution Observations*. Presentation. Ocean Sciences Meeting, New Orleans, LA.

December 8, 2023. *Machine Learning Radiative Transfer Emulation for Satellite Data Assimilation*. Poster. Earth System Science Poster Conference, Boulder, CO.

December 9, 2022. *A Machine Learning Augmented Data Assimilation Method*. Poster. Earth System Science Poster Conference, Boulder, CO.

June 8, 2022. *Augmenting the EnKF with a Shallow Convolutional Neural Network*. Poster. International Symposium on Data Assimilation, Fort Collins, CO.

March 13, 2020. *Uncertainty Quantification and Calibration of a Large Subsurface Flow and Transport model using the Ensemble Kalman Filter*. Waste Management Symposium, Phoenix, AZ.

## Teaching Experience

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|-------------|----------------------------------------------------------------------------------------------|-------------------|-----------------------|
| Fall 2025   | <b>ATOC 4870/5870: Climate Modeling Laboratory</b>                                           | Grader            | CU Boulder            |
| Spring 2025 | <b>ATOC 5060: Dynamics of the Atmosphere and Oceans</b>                                      | Grader            | CU Boulder            |
| Fall 2023   | <b>ATOC 4500/5500: Chaos and Predictability</b>                                              | Grader            | CU Boulder            |
| Spring 2022 | <b>ATOC 1070: Weather and Atmosphere Laboratory</b>                                          | TA/Section Leader | CU Boulder            |
| Summer 2015 | <b>Applied Statistics for Surface Water Hydrology</b> , Co-taught with Professor Donna Rizzo |                   | University of Vermont |

## Service

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## PEER REVIEW

2025-26 **Journal of Advances in Modeling Earth Systems (JAMES)** Reviewer

2025-26 **Monthly Weather Review** Reviewer

## UNIVERSITY OF COLORADO, BOULDER, DEPARTMENT OF ATMOSPHERIC AND OCEANIC SCIENCE

2024/25 **ATOC ML Journal Club** Co-organizer

2024/25 **ATOC Outreach Committee** Member

2024/25 **ATOC Community Committee** Member

2024 **ATOC REU Program** Graduate Student Mentor

2023/24 **Awards Committee**, Member

2023/24 **Space Committee**, Graduate Student Lead

2022/23 **Student Concerns Committee**, Graduate Student Lead

2022/23 **Prospective Student Visit Committee**, Member

2022/23 **Curriculum Committee**, Member

2022/23 **Space Committee**, Member

2022/23 **ATOC Colloquium Committee**, Member

2022/23 **REU Planning Committee**, Member

2021/22 **Student Concerns Committee**, Member

2021 **Graduate Application Mentor Program**, Volunteer Application Mentor

## Other Professional Experience \_\_\_\_\_

### **Hydrogeologic, Inc.**

*Reston, VA*

STAFF ENGINEER/LEADER

*2020-2021*

STAFF ENGINEER

*2018-2020*

ASSOCIATE ENGINEER

*2016-2018*

- Developed and applied groundwater and surface water models to support environmental remediation and planning at federal facilities.
- Led modeling and data analysis projects using ensemble Kalman filter approaches to quantify uncertainty and inform client decision-making.
- Managed technical deliverables and communication with clients and stakeholders, resulting in high-quality reports and minimal revisions.
- Implemented automated workflows and particle tracking simulations to optimize model performance and efficiency.
- Presented key findings at the 2020 Waste Management Symposium, demonstrating practical application of advanced modeling techniques.

### **Hydrogeologic, Inc.**

*Redbank, NJ*

RESEARCH AND DEVELOPMENT INTERN

*Winter 2014/2015*

- Integrated optimization algorithms with hydraulic modeling software to automate flood control design, outperforming expert manual designs.

## Skills \_\_\_\_\_

**Computer Languages:** Python (NumPy, Scipy, Xarray, Pandas, Keras, TensorFlow, Pytorch, PyMC), c++, MATLAB, Octave, Mathematica, VisualBasic

**Software:** Git, JEDI, ArcGIS, QGIS, VSCode

**HPC:** Derecho/Casper, CURC, Slurm, PBS, Spack, CUDA